# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-VEC3-EN · v2.2 · 2026-11-02 |
| Package | star-vec3 |
| Version | 0.1.0 |
| Capability | star.geometry.vec3 |
| Maturity | Released & Verified (RELEASED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_vec3 |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_vec3-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_vec3-0.1.0-py3-none-any.whl | 1128bbe0700120156a19a5578cd39ce6c5b82c1f39c078fa527b62444ce7b6fc |
| star_vec3.py | 6fb84ab460f69ff4d86e203000ef517d426a57288cbc10432a4b8e9c9ed71a00 |
| star_vec3-0.1.0-source.zip | d489d632bb6c3f087b760839bc0081a44faebd9059afe7d9ac611ec8024031b8 |

# 5. What it does, and its limits

Dot, cross and triple product, length, distance, unit vector, accurate angle between directions, projection and rejection of three-dimensional vectors; within 6e-16 of SPICE and ERFA and 8e-15 of NumPy on 600 triples of vectors, nearly parallel pairs included; malformed vectors and zero directions refused.

No matrices, no rotations (see `star_quaternion` and `star_euler`), no vectors of other dimensions. The triple product is compared with one library (NumPy) only. Products of components below 1e-154 underflow to zero in `dot`, `cross` and `triple` as in any double-precision code; `norm`, `unit` and `angle_deg` are not affected.

- `dot`, `cross`, `triple`, `norm`, `distance`, `unit`: the definitions, with the length computed without intermediate overflow or underflow and the sums compensated.
- `angle_deg` as atan2(|a × b|, a · b) on unit vectors: in [0, 180], accurate near 0 and near 180 degrees and independent of the lengths; `project` and `reject`, which add up to the first vector.
- Measured on 600 triples of vectors (lengths 1e-3 to 1e9; 200 pairs within 1e-9 to 1e-3 rad of parallel or antiparallel): within 6e-16 of SPICE and ERFA relative to the lengths involved, angles within 5e-14 deg; the triple product within 8e-15 of NumPy's determinant.
- Every function returns floats or raises `ValueError`: a vector that is not a list or tuple of exactly three finite real numbers within ±1e100 (booleans refused), or the zero vector where a direction is needed.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before

using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_vec3_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-vec3}: The three-dimensional vector operations of orbit and attitude code},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_vec3/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts